

Entry survey CS 101 (Check-In)

This short survey helps me understand your background, tools, and expectations so I can better support you in this course. After the survey, you will watch a short video and answer a few questions to test your understanding. **Incentive:** Extra credit for completion by **Friday, Jan 30, 5:00 PM PST**. Please provide your email (for my eyes only) to get the extra credit. Otherwise the survey answers are completely anonymous. **Disclaimer:** This survey was created with assistance from an AI tool (ChatGPT 5.2) to help draft and organize questions. The final content and use of responses are determined by the instructor.

* Indicates required question

1. Email *

2. How would you currently assess your programming skills (in the programming language you know best)? *

Mark only one oval.

- ☐ Beginner (still learning basic syntax and core concepts)
- ☐ Early intermediate (can write small programs with guidance)
- ☐ Intermediate (can design, implement, and debug programs independently)
- ☐ Advanced (experience with multiple languages on larger projects)

3. Which programming languages have you used before? (Check all that apply) *

Check all that apply.

- ☐ Java (from CS 100)
- ☐ C
- ☐ C++
- ☐ Python
- ☐ JavaScript
- ☐ R
- ☐ Other (please specify)

4. What operating system does the computer that you use for this course run? *

Mark only one oval.

- ☐ Windows
- ☐ macOS
- ☐ Linux
- ☐ Not sure

5. Have you ever used a command line / terminal? *

Mark only one oval.

- ☐ No
- ☐ Yes, a little
- ☐ Yes, regularly

6. If yes, where have you used it? (Check all that apply)

Check all that apply.

- ☐ Windows Command Prompt / PowerShell
- ☐ macOS Terminal
- ☐ Linux shell (bash/zsh/etc.)
- ☐ Git
- ☐ Remote server (ssh)
- ☐ Other: _____

7. Have you ever compiled a program from the command line?

Mark only one oval.

- ☐ Yes
- ☐ No
- ☐ Not sure

8. Realistically, how much time can you devote to this course *outside of class* per week? *

Mark only one oval.

- ☐ Less than 2 hours
- ☐ 2–4 hours
- ☐ 4–6 hours
- ☐ 6+ hours

9. Do you expect any constraints this term that might affect your participation? *

10. What is your motivation for taking this course? Check all that apply. *

Check all that apply.

- ☐ Degree requirement
- ☐ Interest in systems / low-level programming
- ☐ Career preparation
- ☐ Curiosity
- ☐ Other (short answer)

11. What do you most want to get out of this course? *

12. Have you used AI tools (e.g., ChatGPT, Copilot, Claude) for programming? *

Mark only one oval.

- ☐ No
- ☐ Yes, occasionally
- ☐ Yes, regularly

13. If yes, what have you used them for? (Check all that apply)

Check all that apply.

- ☐ Explaining code
- ☐ Debugging
- ☐ Writing small programs
- ☐ Learning syntax
- ☐ Non-programming tasks
- ☐ Other: _____

Quiz

In this section, you're given a short video and asked a few questions to test your understanding. The subject of the video is [the programming language C++](#), which will be our focus in CS 101, which is "*infamous for its steep learning curve*" (according to the video). This is true and it wouldn't be my choice for a first language, but it is excellent for learning data structures, getting you warmed up, for algorithms. For fun, you can compare it with "[Java in 100 seconds](#)" by the same guy (I don't know Java well at all, by the way, so you're ahead of me!).

Watch the video (3 min) and test your understanding by answering a few questions below. Source: Fireship, Feb 7, 2022 ([YouTube](https://www.youtube.com/watch?v=MNeX4EGtR5Y)).



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14. How is C++ best described in terms of typing and execution? *

Mark only one oval.

- ☐ A dynamically typed, interpreted language
- ☐ A scripting language that runs in a virtual machine
- ☐ A markup-based configuration language
- ☐ A statically typed, compiled programming language

15. What is the relationship between C and C++ as described in the video? *

Mark only one oval.

- ☐ C++ replaces C entirely
- ☐ Virtually any C program is also a valid C++ program
- ☐ C and C++ are unrelated languages
- ☐ C++ is interpreted on top of C

16. Why is C++ widely used in systems with constrained resources? *

Mark only one oval.

- ☐ It automatically manages all memory for the programmer
- ☐ It is easier to learn than most languages
- ☐ It provides low-level memory and hardware control with high-level abstractions
- ☐ It does not require compilation

17. In C++, what is a class? *

Mark only one oval.

- ☐ A blueprint for creating objects with attributes and methods
- ☐ A special function used to manage memory
- ☐ A built-in container type for strings
- ☐ A compiler directive

18. Which of the following is **not** a core object-oriented concept supported by C++? *

Mark only one oval.

- ☐ Encapsulation
- ☐ Polymorphism
- ☐ Inheritance
- ☐ Automatic garbage collection

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